

TM Forum Introductory Guide

Autonomous Networks Framework

IG1218F

Maturity Level: General Availability (GA)	Team Approved Date: 14-Mar-2025
Release Status: Pre-production	Approval Status: TM Forum Approved
Version 2.0.0	IPR Mode: RAND

Notice

Copyright © TM Forum 2025. All Rights Reserved.

This document and translations of it may be copied and furnished to others, and derivative works that comment on or otherwise explain it or assist in its implementation may be prepared, copied, published, and distributed, in whole or in part, without restriction of any kind, provided that the above copyright notice and this section are included on all such copies and derivative works. However, this document itself may not be modified in any way, including by removing the copyright notice or references to TM FORUM, except as needed for the purpose of developing any document or deliverable produced by a TM FORUM Collaboration Project Team (in which case the rules applicable to copyrights, as set forth in the [TM FORUM IPR Policy](#), must be followed) or as required to translate it into languages other than English.

The limited permissions granted above are perpetual and will not be revoked by TM FORUM or its successors or assigns.

This document and the information contained herein is provided on an “AS IS” basis and TM FORUM DISCLAIMS ALL WARRANTIES, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY WARRANTY THAT THE USE OF THE INFORMATION HEREIN WILL NOT INFRINGE ANY OWNERSHIP RIGHTS OR ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE.

TM FORUM invites any TM FORUM Member or any other party that believes it has patent claims that would necessarily be infringed by implementations of this TM Forum Standards Final Deliverable, to notify the TM FORUM Team Administrator and provide an indication of its willingness to grant patent licenses to such patent claims in a manner consistent with the IPR Mode of the TM FORUM Collaboration Project Team that produced this deliverable.

The TM FORUM invites any party to contact the TM FORUM Team Administrator if it is aware of a claim of ownership of any patent claims that would necessarily be infringed by implementations of this TM FORUM Standards Final Deliverable by a patent holder that is not willing to provide a license to such patent claims in a manner consistent with the IPR Mode of the TM FORUM Collaboration Project Team that produced this TM FORUM Standards Final Deliverable. TM FORUM may include such claims on its website but disclaims any obligation to do so.

TM FORUM takes no position regarding the validity or scope of any intellectual property or other rights that might be claimed to pertain to the implementation or use of the technology described in this TM FORUM Standards Final Deliverable or the extent to which any license under such rights might or might not be available; neither does it represent that it has made any effort to identify any such rights. Information on TM FORUM's procedures with respect to rights in any document or deliverable produced by a TM FORUM Collaboration Project Team can be found on the TM FORUM website. Copies of claims of rights made available for publication and any assurances of licenses to be made available, or the result of an attempt made to obtain a general license or permission for the use of such proprietary rights by implementers or users of this TM FORUM Standards Final Deliverable, can be obtained from the TM FORUM Team Administrator. TM FORUM makes no representation that any information or list of intellectual property rights will at any time be complete, or that any claims in such list are, in fact, Essential Claims.

Direct inquiries to the TM Forum office:

181 New Road, Suite 304
Parsippany, NJ 07054, USA
Tel No. +1 862 227 1648
TM Forum Web Page: www.tmforum.org

Table of Contents

Notice	2
Table of Contents	4
Executive Summary.....	5
Introduction.....	6
1. The Autonomous Networks Framework Overview	7
2. Key Elements of AN Framework.....	8
2.1. Key Effectiveness Indicators	8
2.1.1. Indicator Framework	8
2.1.2. KEI Reference Set	10
2.2. Autonomous Network Levels	10
2.2.1. Generic AN L3/L4 characteristics	10
2.2.2. Technical Domain-specific AN L3/L4 characteristics	12
2.2.3. AN Level Taxonomy.....	14
2.3. AN Target Architecture	17
2.3.1. Conceptual Architecture.....	17
2.3.2. AN Reference Architecture	19
2.3.3. AI in the AN L4 Reference Architecture.....	19
2.4. AN Map / High-Value Scenarios.....	20
2.4.1. High-Value Scenarios and AN Scenario Mapping	21
2.4.2. Prioritize the Scenarios	22
3. Autonomous Networks Practice.....	24
4. Terms & Abbreviations Used within this Document	25
4.1. Terminology	25
5. References.....	26
6. Administrative Appendix	27
6.1. Document History	27
6.1.1. Version History.....	27
6.1.2. Release History.....	27
6.2. Acknowledgments.....	28

Executive Summary

The AN Framework provides CSPs with a structured methodology and a comprehensive set of tools for navigating their AN transformation journey, including guidance and support for achieving the Level 4 goals of Zero-X capabilities and Self-X experiences (outlined in IG1218).

This framework consists of the vision and implementation methodology, which includes foundational elements, the operational practices, and industry assessment and certification. The 4 foundational elements of AN Framework are:

1. **Key Effectiveness Indicators.** KEIs help CSPs identify what benefits they could receive by upgrading their telecommunications system with more autonomy capabilities.
2. **AN Levels.** Operators benefit from having a clear view on their expected maturity of their network so that they can focus on the important features they should prioritize to achieve a defined level of autonomy in their networks.
3. **AN Target Architecture.** The realizations of zero-touch operations using autonomous network reference architectures aims to focus solutions on the essential principles and patterns that will improve Self-X capabilities and thereby help operators to reach L3 and L4 levels of autonomy.
4. **Scenarios (or "AN Map").** A mapping of high-value scenarios that helps CSPs define, prioritize, and align autonomous network capabilities with key operational and business objectives across different network lifecycle phases and service domains. It is designed to help CSPs set and decompose overall AN evolution objective from an end-to-end perspective and specify the direction and priority of *capability* development.

These four elements, while relatively independent, need to be further organized to form a pragmatic how-to guide to facilitate implementation of AN. For each of these elements, pragmatic 'how-to' suggestions, as well as reference to applicable standards, are provided in this document. This document also introduces the operational practice, covering two concepts, AN strategic planning and AN Journey. The industry assessment and certification of AN is at an earlier stage of development, and will be elaborated in a future revision of this document.

Introduction

This document introduces the overall framework and practice methods of AN Framework (ANF). It aims to provide a systematic method for operators from AN strategy planning to implementation, and guide operators to carry out AN work more efficiently. C-level and senior managers, should focus on the value proposition and AN level characteristics of AN Framework, control the direction of value and capability evolution of AN, understand the value brought by AN, and confirm whether AN goals are achieved. The general owner of the AN work can focus on the ANF's effectiveness indicators, evaluation standards, target architecture, and the AN map, to formulate the operator's strategic plan (ANSP), specify the overall objective and evolution path, define the key business scope of the AN, and coordinate the work division and capability evolution pace of each system. The AN implementation personnel of various technology and service domains can focus on the AN Journey, master the closed-loop implementation methods based on value scenarios, and perform iterative evolution to continuously improve the AN level.

1. The Autonomous Networks Framework Overview

The AN Framework is an out-of-the-box implementation guide and toolkit to guide CSPs' AN transformation so that they can plan and implement AN in a more efficient and systematic manner and more quickly fulfill the vision of Zero-X and Self-X.

The Autonomous Network Framework (ANF) is structured into two key aspects:

1. **AN Foundations** (depicted in the top half of the figure 1) – These are the core building blocks that define the vision, methodology, and essential elements needed to enable AN evolution. They include **Key Effectiveness Indicators (KEIs)**, **AN Levels**, **the AN Target Architecture**, and **the "AN Map"** or High-Value Scenarios matrix. These four elements provide the structure for AN development.
2. **AN Practices** (depicted in the bottom half of the figure 1) – These focus on putting AN into action, guiding CSPs through the implementation and operationalization of AN. AN practices consist of two key activities:
 - **AN Strategic Planning:** A structured approach for CSPs to define and guide their AN evolution. It includes corporate-level AN strategies, executive commitments, and guidance on implementing the four foundational elements of ANF. Additionally, it outlines industry contributions and technological innovation directions.
 - **The AN Journey:** An **iterative process** where operators execute strategies, measure progress, and refine their approach to achieve AN objectives.

By leveraging the AN Foundations within operator AN Practices, CSPs can systematically evolve their networks toward higher levels of autonomy, ensuring both strategic alignment and practical execution.

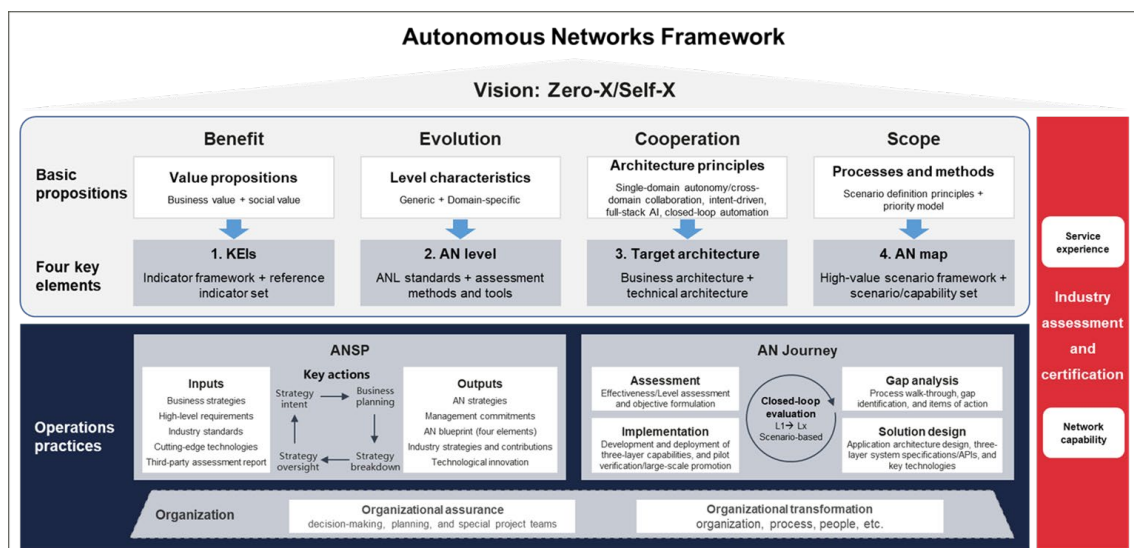


Figure 1. Autonomous Networks Framework

The AN Framework includes all four key elements, operational practices, training, assessment, and certification.

2. Key Elements of AN Framework

2.1. Key Effectiveness Indicators

When an enterprise plans to undertake a strategy or investment, it must align its values and goals within the organization and measure the return on investment. Central to this process is the definition and measurement of key indicators.

IG1252 [3] defines an evaluation methodology for Autonomous Networks Levels (AN Level). AN Levels gauge the level of autonomy capability in a given operational workflow or autonomous domain. However, AN Levels do not adequately reflect the overall impact or value of an autonomous network. Hence, there is a need to establish Key Effectiveness Indicators (KEIs) to help CSPs in determining the benefits of enhancing their telecommunications system with greater autonomy capabilities.

The value of these indicators lies in two aspects:

1. visualizing and quantifying the effectiveness and benefits of AN evolution,
2. and aligning the development of autonomous capabilities with the enterprise strategy and service development trend.

Effectiveness indicators and AN levels are the two key AN evaluation factors that jointly facilitate the fulfillment of the Autonomous Network vision. KEIs can be used in the following three situations:

1. Formulating *improvement objectives* for AN based on KEIs. For example, if the service delivery target at L4 is measured in minutes, the target baseline can be set based on an opco's O&M status, experience of benchmark CSPs and annual investment budget. Note that the baseline value is not equivalent to a characteristic of L4.
2. Guiding *capability development* by associating KEIs with operation capabilities and converting and breaking down capability requirements for example, the time needed for optical transport network private line service delivery can be broken down into various key capability indicators, such as the time needed for CPE deployment and solution design. These capability metrics can be mapped to specific functional requirements for support systems, operations and maintenance (O&M) centers, and network elements.
3. Verifying the *effectiveness of operation capability* development based on KEIs to provide inputs for the next AN Journey – the inputs include adjustments to the rationality of objectives, investment plan changes and optimized development directions.

2.1.1. Indicator Framework

Traditional KPIs are typically established in response to O&M challenges or identified weaknesses. For instance, when there are numerous customer complaints, indicators like the complaint rate are introduced to guide teams in enhancing product quality and service processes while reducing customer grievances. However, this approach primarily addresses isolated issues and lacks comprehensive planning and target guidance. It falls short of systematically steering organizational and platform capability development. In light of this, drawing from the balanced scorecard theory, the existing **TM Forum Metrics** framework (GB935 [10] / GB988 [11], etc.), and insights gained from operator autonomous network KEI design practices, we propose

an autonomous network KEI framework. The design and decomposition methodology for this framework are illustrated in the following diagram, Figure 2.

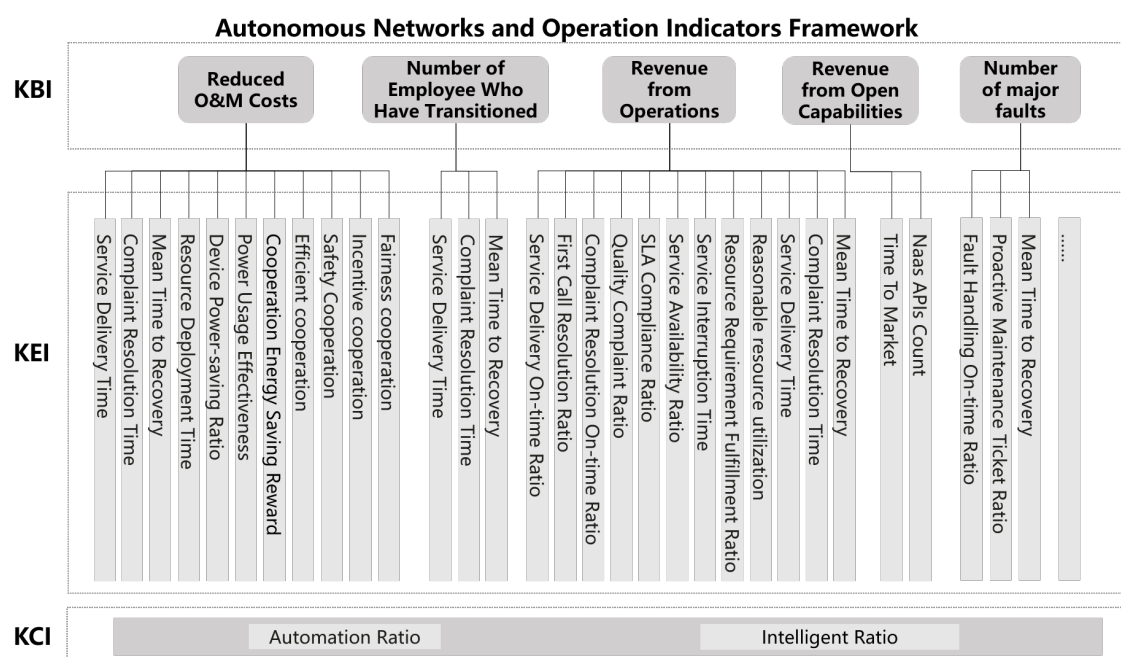


Figure 2. Indicator Framework

The key features of this indicator framework are outlined below:

1. **Value-Centric Approach:** KEIs are deliberately designed from a 'top-down' perspective, aligning with the four primary value propositions of the autonomous network. Each KEI is directly or indirectly linked to enhancing one or more of these value propositions. A corresponding set of effectiveness indicators is established for each recognized value proposition, facilitating effective measurement of the value delivered.
2. **Multi-Level Hierarchical Structure:** The KEIs are structured hierarchically, systematically breaking down the measurement levels based on the identified value propositions. This hierarchical structure forms a causal indicator tree. The KPIs reflect the main AN value propositions, offering a holistic view of AN's impact at both the company and industry levels. Simultaneously, it provides guidance on achieving effectiveness and establishes internal appraisal requirements. Within each domain, detailed KPI rules can be formulated, and specific targets can be set based on the domain's unique KPIs and specialized areas of business. Furthermore, **KCIs** can be further subdivided, driven by an analysis of process activities and identified capability gaps during the AN journey within valuable scenarios. This sub-level decomposition guides the development of system capabilities more effectively.

The distinction between KPIs and KCIs lies in their respective purposes. KPIs are business-oriented KEIs that focus on end-to-end outcomes, while KCIs serve as capability measures aimed at bolstering the attainment of Key Effectiveness Indicators (KEIs) by addressing specific process activity weaknesses. **In essence, KEIs measure the achievement of business objectives, while KCIs gauge key capabilities.** As technology and capabilities evolve, the essential KCIs within a process may undergo changes, whereas KEIs tend to remain relatively stable. For instance, consider a troubleshooting scenario within a domain where the KEI is the average troubleshooting duration. At

present, accurately identifying faults poses a challenge. Therefore, a suitable KCI can be defined as 'Fault Identification Accuracy.' As fault identification capabilities improve, the key weakness may shift to fault diagnosis. Accordingly, KCIs can be redefined as 'Automatic Fault Diagnosis Rate' and 'Fault Diagnosis Accuracy Rate' to address this evolving capability requirement.

3. **Consider both benefit and efficiency:** When defining KEIs, it is important to consider not only efficiency indicators like automation rates but also business benefit indicators such as time consumption and quality. By incorporating both benefit and efficiency aspects, a more comprehensive measurement system for KEIs can be developed. For example, in troubleshooting scenarios, it's essential to define not just the efficiency indicator of troubleshooting automation rates but also the benefit indicator of average troubleshooting duration.

2.1.2. KEI Reference Set

For details about the reference set of effectiveness indicators and the definition of specific indicators, see IG1256 Autonomous Network Effectiveness Indicators [4].

The following is an example:

Indicator name	Time to Market	Abbreviation	TTM
		Unit	Day
Definition of the indicator	Measure the time taken for new services or products from the initial concept stage to market launch.		
Indicator Description	This indicator reflects the ability of enterprises to launch new services or products. The shorter the time to market, the quicker a return on investment can be realized. AN capability such as self-orchestration and self-loading can quickly support the design, development, verification, and release of new services and products. The intent-based interface simplifies inter-system interconnection and shortens the integration time. All these AN capabilities can effectively shorten the product launch time.		
Calculation formula	TTM = Total time from concept to launch of all new services or products ÷ Number of new services or products Go-to-market (or go-to-shelf) refers to the formal sale of a product in the market and users can already buy the product.		

Table 1. KEI Reference Set Example

2.2. Autonomous Network Levels

2.2.1. Generic AN L3/L4 characteristics

There is now industry consensus that AN Levels are defined based on cognitive closed-loop theory and the degrees of a human-machine division of labor. An increasing number of AN practitioners want to take an approach that is straightforward in that it defines AN Level characteristics in terms of business or service value. The IG1326 Autonomous Networks: Empowering digital transformation – evolving from Level 2/3 towards Level 4 [9] has summarized general L4 characteristics and the key points of L3-to-L4 evolution from business and capability perspectives as shown in Table 2 below. Domain-specific L4 characteristics can be defined based on the guiding principles of the general L4 characteristics shown in the table.

Generic L3/L4 Characteristic				
Perspective	Dimension	L3 Characteristic	L4 Characteristic	Description
Customer	Zero-wait	Automated service provisioning	Automated service delivery	L3: Services are configured automatically in the provisioning phase. Other phases involve manual operations. L4: Resource surveys are no longer required. Self-acceptance is supported. E2E automation is implemented from receiving delivery tickets by network personnel to returning the tickets.
	Zero-trouble	Experience awareness and visualization	Experience evaluation and assurance	L3: User experience is visualized, and complaints are diagnosed automatically. L4: Experience can be evaluated and guaranteed, and warnings can be generated for potential complaints.
	Zero-touch	Visualization	Interaction	L3: Customer service provisioning progress and SLA are visualized. L4: Proactive recommendation and intelligent response are supported.
Network	Self-configuration	Automated configuration delivery	Pre-event simulation Post-event verification	L3: Configurations are automatically delivered and manually reviewed.

Generic L3/L4 Characteristic				
				L4: Configurations are automatically generated and simulated online, eliminating the need for manual review.
	Self-healing	Precise fault diagnosis	Potential risk prediction and prevention	L3: Faults can be accurately demarcated and located. L4: Potential risks can be predicted and prevented, and faults can be automatically rectified.
	Self-optimizing	Single-objective exclusive optimization	Multi-objective collaborative optimization	L3: Optimization can be performed exclusively for a single objective – for example, wireless network performance. L4: Optimization can be performed for multiple objectives – for example, user experience and energy efficiency.

Table 2. Evolving from L3 to L4

2.2.2. Technical Domain-specific AN L3/L4 characteristics

Based on the generic AN L3/L4 characteristics, the technical domain AN L3/L4 characteristics are summarized as follows:

Wireless Network L3/L4 Characteristic		
Dimension	L3 Characteristic	L4 Characteristic
Intelligent Sensing	Non-real-time sensing	Real-time intelligent sensing
Intelligent and Simplified O&M	Machine-assisted O&M, precise diagnosis	Machine-guided O&M, intent driven prediction and prevention
Intelligent Network Optimization	Single-objective optimization	Intent driven multi-Objective optimization
Intelligent Service Operation	Best Effort Experience	Intent driven deterministic Experience

Table 3. Wireless Network Evolving from L3 to L4

Core Network L3/L4 Characteristic		
Dimension	L3 Characteristic	L4 Characteristic
Intelligent O&M	Rule-based automation in some scenarios	Automation in most scenarios based on AI technologies
Agile Delivery	Rule-based provisioning, change, verification and testing, some scenarios	AI-based check, AI and digital twin-based verification and testing, cover almost all scenarios
High Reliability	Fast recovery from multipoint faults	Proactive prevention, intelligent risk prevention and control

Table 4. Core Network Evolving from L3 to L4

IP Network L3/L4 Characteristic		
Dimension	L3 Characteristic	L4 Characteristic
Agile Service Routing	Work order-driven, day-level provisioning	Application driven, minute-Level intent provisioning
Premium Deterministic Experience	Optimal latency and self-tuning	Deterministic experience, guaranteed latency and availability, and proactive self-tuning
No Config Error	Resource-based autoconfiguration	Service intent-based configuration with pre-check and post-check validation
Efficiency Fault-removing	Fault locating and isolation with on-path flow-based OAM mechanism	Intelligent fault/potential risk identification and self-demarcation and locating in minutes

Table 5. IP Network Evolving from L3 to L4

Access Network L3/L4 Characteristic		
Dimension	L3 Characteristic	L4 Characteristic
Online Service Deployment	Work order-driven one-stop planning, construction, and verification, enabling the system in days	Application-driven, application and flow level deterministic experience slicing, enabling services in seconds
Customer Complaint Handling	Broadband experience visualization	Experience evaluation and self-repair
Quality Optimization	Proactive identification of experience faults and hour-level fault locating and demarcation	Proactive identification and demarcation, and self-optimization

Table 6. Access Network Evolving from L3 to L4

Transmission Network L3/L4 Characteristic		
Dimension	L3 Characteristic	L4 Characteristic
Assorted Service Automation	Work order-driven, day-level provisioning	Compute/Cloud application driven, online wavelength planning, minute-level provisioning
Ultimate SLA Guaranteed	Optimal latency, automatic SLA configuration	Deterministic bandwidth + latency + availability commitment, interactive SLA self-tuning
Time-saving Ticket Journey	Static rule filtering, hour-level demarcation and locating	Intelligent fault/potential risk identification and interactive self-demarcation and locating in minutes

Table 7. Transmission Network Evolving from L3 to L4

2.2.3. AN Level Taxonomy

The AN Level Taxonomy classifies the degree or *level* of automation and intelligence in CSP networks, ranging from basic automation to full autonomy. This standard can be used to evaluate the levels of autonomy, identify weaknesses in autonomous capabilities, analyze gaps between the status quo and objectives, and guide CSPs to systematically plan and deploy autonomous capabilities.

The technical work of AN Level evaluation methodology is published in IG1252 Autonomous Network Levels Evaluation Methodology [3], which describes the AN Level evaluation approach and operation flows, tasks evaluation criteria, scoring method, etc. The table below shows the levels in more detail, with additional explanation of the levels following.

Autonomous Levels	L0: Manual Operation & Maintenance	L1: Assisted Operation & Maintenance	L2: Partial Autonomous Networks	L3: Conditional Autonomous Networks	L4: Highly Autonomous Networks	L5: Fully Autonomous Networks
Execution	P	P/S	S	S	S	S
Awareness	P	P/S	P/S	S	S	S
Analysis	P	P	P/S	P/S	S	S
Decision	P	P	P	P/S	S	S
Intent / Experience	P	P	P	P	P/S	S
Applicability	N/A	Selected Scenarios				All Scenarios



People (manual)



System (autonomous)

Table 8. Autonomous Network Level Taxonomy

Level 0 – Manual management: the system delivers assisted monitoring capabilities, which means all dynamic tasks must be executed manually.

Level 1 – Assisted management: the system executes a certain repetitive sub-task based on pre-configuration to increase execution efficiency.

Level 2 – Partially Autonomous Networks: the system enables partial automatic O&M for certain units based on pre-defined rules or policies under certain external environments.

Level 3 – Conditionally Autonomous Networks: building on L2 capabilities, the system with awareness can sense real-time environmental changes, and in certain network domains, optimize and adjust itself to the external environment.

Level 4 – Highly Autonomous Networks: building on L3 capabilities, the system enables, in a more complicated cross-domain environment, analysis and decision-making based on predictive or active closed-loop management of service and customer experience-driven networks.

Level 5 – Fully Autonomous Networks: this level is the goal for telecom network evolution. The system possesses closed-loop automation capabilities across multiple services, multiple domains and the entire lifecycle, achieving Autonomous Networks.

The AN Level can be applied to generic domain or to a specific network domain (i.e domain-specific). On the one hand, we need to unify the concepts, principles and methods so that the AN Levels of different networks and services can be compared horizontally. On the other hand, the specific network and service characteristics of a given domain must be considered to better guide the AN capability planning and construction of different networks and services.

The IG1305 Autonomous Networks: Empowering digital transformation – from strategy to implementation [8] further describes the generic AN Level and domain-specific AN Level and the relationship between them.

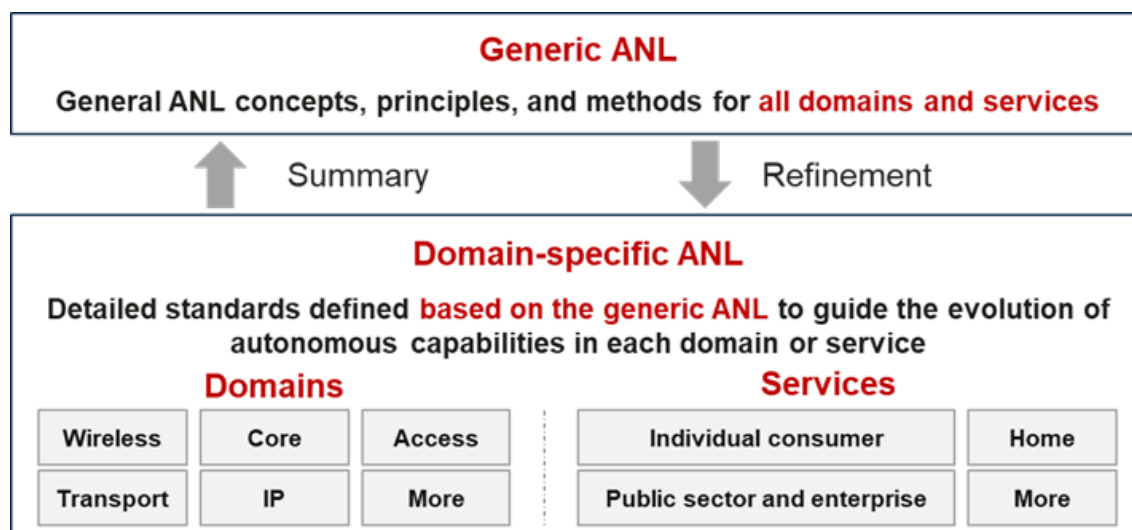


Figure 3. Generic AN Level Characteristics and Domain-specific AN Level Characteristics

- **Generic AN Level** is applicable to all domains and services. It focuses on the AN Level concepts, principles, methods, and O&M task baselines, offering basic categories for the usage and instantiation of domain-specific AN Level. The core value of generic AN Level lies in ensuring consistent dimensions, granularities, and terms of O&M tasks between different domains or services to support horizontal comparison and analysis of the AN Level evaluation result. The Generic AN Level definition and evaluation method are described in TM Forum IG1252 [3].
- **Domain-specific AN Level** is applicable to a specific domain or service. It defines the detailed rules for the domain or service based on the generic AN Level, including the assessment rules of L1 to L5 and baseline scenarios. Domain-specific AN Level can be used to evaluate the AN Level of the domain or service and guide the evolution of autonomous capabilities in the domain or service. For templates and examples of domain-specific AN Level, see Figure 4. Domain-specific AN Level and evaluation method are defined by respective SDOs, e.g. 3GPP network and service AN Level is defined in 3GPP. TS28.100 [7].

Domain-specific ANL						Function breakdown		
Operation flow	Sub-operation flow	Operation task	Evaluation Criteria (XX Domain)	Selected Scenario		Function Map		
				Sub-scenario1	Sub-scenario2	BSS	OSS	Network Layer
Maintenance	Fault management	Fault identification	L1: L2: L3: L4:	2	3	N/A	L3: L4:	L3: L4:
		Demarcation & Location	...			...	...	...

Figure 4. Domain-specific AN Level and function map

To systematically strategize and cultivate autonomous capabilities while providing clearer guidance to Operators in reaching Levels 3 (L3) and 4 (L4), we deconstruct functional requirements within the BSS, OSS, and network layers (comprising NMSs and NEs). This deconstruction is based on domain-specific Autonomous Network Level criteria and the key AN architecture principles, namely single-domain autonomy and cross-domain collaboration. The resulting functional requirements collectively compose what we refer to as the 'function map.' The domain-specific AN Level defines the objectives of L3 and L4, whereas the function map illustrates the means to achieve these levels. Figure 4 provides an illustrative example of domain-specific AN Level, with a specific focus on the function map related to the domain-specific AN Level.

CSPs need to determine whether they require a comprehensive and systematic evaluation or if a preliminary, more cursory assessment suffices. Systematic evaluation involves a comprehensive assessment of the autonomy level within a specific domain based on domain-specific AN Level criteria. It entails comparing the existing state to predefined objectives and pinpointing any identified weaknesses and gaps. The function map serves as a valuable tool for strategizing and evolving system functions aimed at enhancing weaknesses and closing gaps. For a rapid and preliminary assessment, a questionnaire is devised, based on the domain-specific AN Level. This questionnaire enables participants to swiftly grasp the evaluation process and provide a rough estimation of a Communication Service Provider's (CSP) autonomy level.

2.3. AN Target Architecture

The complete AN target architecture includes the architecture vision, AN framework, architecture principles and technical reference architecture comprehensively defined by TM Forum, ETSI and domain-specific SDOs.

To define the Autonomous Network (AN) target architecture, CSPs can follow this structured process:

1. **Establish the architectural vision and principles** by drawing insights from corporate strategic directives, business strategies, technical prerequisites, architecture requirements, AN standards, and prevailing industry technology trends.
2. **Articulate a reference architecture** that aligns with corporate strategic goals, drawing from AN's functional reference architecture and existing architectural capabilities. This involves conducting process walk-throughs, identifying value streams, and summarizing business capability requirements. Throughout this phase, CSPs should identify significant architecture gaps and provide essential improvement recommendations. Operators should particularly emphasize two key aspects: defining an efficient architectural division of labor to expedite service innovation and reduce overall costs, and implementing formalized data and models to support hierarchical AI integration and closed-loop automation.
3. **Put architectural principles into practice** in alignment with corporate strategic objectives for specific business scenarios. Enhance business capabilities based on assessments of AN Level and effectiveness.
4. **Continuously refine the target architecture** in an iterative manner, drawing insights from issues encountered during implementation.

2.3.1. Conceptual Architecture

The TM Forum framework that members are building identifies three layers and four closed loops to deliver Autonomous Networks (see Figure 5). The three layers are common capabilities of operations that can be utilized to support all scenarios and business needs:

1. **Resource Operations layer** – provides automation of network resources and capabilities in each autonomous domain level
2. **Service Operations layer** – provides the capabilities for IT services and network planning, design, rollout, provisioning, assurance and optimization of operations across multiple autonomous domains
3. **Business Operations layer** – provides the capabilities for customer, ecosystem and partner business enabling and operations for AN services

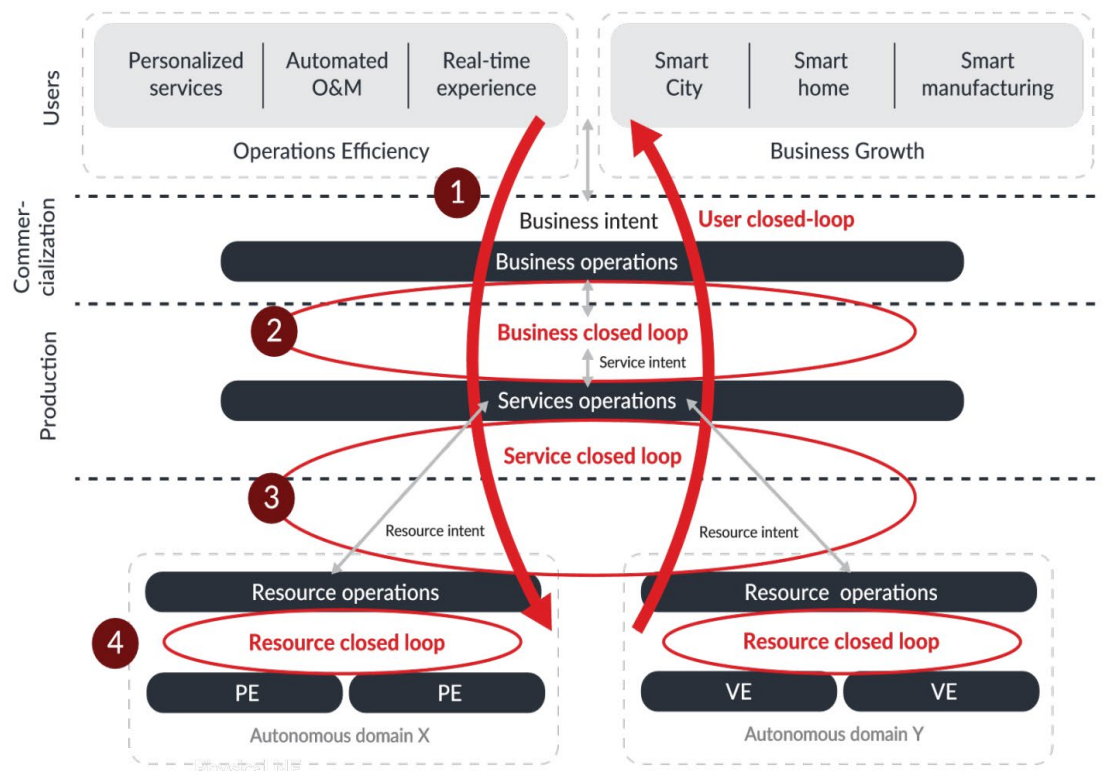


Figure 5. Conceptual Architecture

The four closed loops identified to fulfill the full lifecycle of the inter-layer interaction are:

1. **User closed loop** – the interaction across three layers and three closed loops to support fulfillment of the user's service
2. **Business closed loop** – the interaction between business and service operations may trigger related service and resource closed loops in its fulfillment
3. **Service closed loop** – the interaction between service and network resource operations may trigger related resource closed loops in its fulfillment
4. **Resource closed loop** – the interaction of network resource operations is in the granularity of autonomous domains

Autonomous Networks are characterized by autonomous domains and automated intelligent business, service and resource operations for the closed-loop of digital business, which offer the best-possible user experience, full lifecycle operations automation/autonomy and maximum resource utilization.

The basic principles of the operations of autonomous domains are:

- **Single Domain Autonomy** – each autonomous domain runs in self-operating mode per business objective and hides the details of domain implementation, operations and the functions of the domain elements from the users of the autonomous domains by using an abstraction layer of service APIs.
- **Cross Domain Collaboration** – multiple instances of autonomous domains can be collaborated on by upper layer service operations using the intent-driven interaction to fulfill the lifecycle of network/ICT services.

2.3.2. AN Reference Architecture

Autonomous networks aim to make processes more efficient and intelligent to better serve human needs and enhance overall decision-making. Achieving end-to-end autonomy requires collaboration across the industry. To handle the complexity of various services and technologies, TM Forum and other organizations have created standards and workflows for seamless communication between different layers. TM Forum's autonomous network reference architecture focuses on essential principles and patterns to improve Self-X capabilities, helping operators reach Levels 3 and 4 of autonomy (IG1251 Autonomous Networks – Reference Architecture [2], Figure 6.)

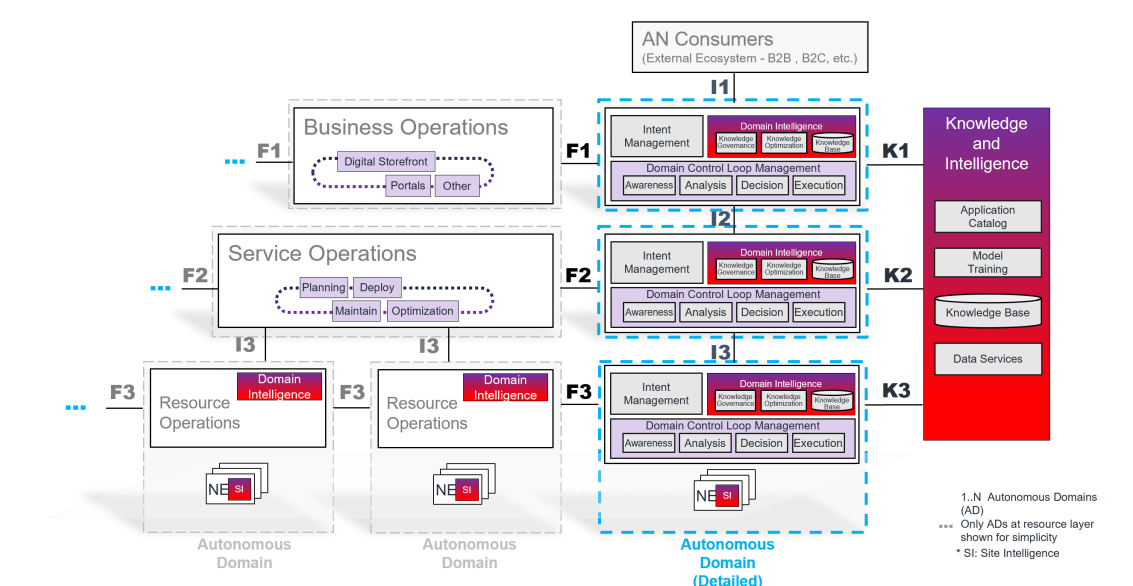


Figure 6. AN Reference Architecture (showing detailed ADs at all layers)

TM Forum IG1251A [5] also presents the details of how technology specific SDOs coordinate with each other on autonomous network architecture. The realizations build on the core concepts of AN (defining closed loops, using intent-driven interfaces) enables the autonomy at various management scopes. Multiple realizations of different management scopes are coordinated with each other and form an end-to-end AN solution to achieve the overall AN vision.

- Domain management: e.g., 3GPP SA5, ETSI F5G, IETF.
- Cross-domain management: e.g., TM Forum AN, ETSI ZSM.

2.3.3. AI in the AN L4 Reference Architecture

At Autonomous Networks Level 4, AI and **Generative AI (GenAI)** are widely integrated into AN's three-layer architecture. Two types of AI-driven agents enhance autonomy at each layer: **copilots** designed for operational roles and **scenario-based agents** tailored for specific operational needs.

- **Business/Service operations:** Agents are developed for users, businesses, and service operations to achieve an E2E autonomous closed loop. AI/GenAI on which the agents depend can be implemented in either of the following modes:
 - Model inference integrated to the agents
 - Model training/inference centrally provided by the cloud AI center

- **Resource operations:** Agents integrating AI/GenAI model inference collaborate with intelligent NEs to implement a closed loop of single-domain network autonomy.
- **Network Elements (NEs):** Built-in AI models offer local inference, high-precision awareness, and control capabilities.

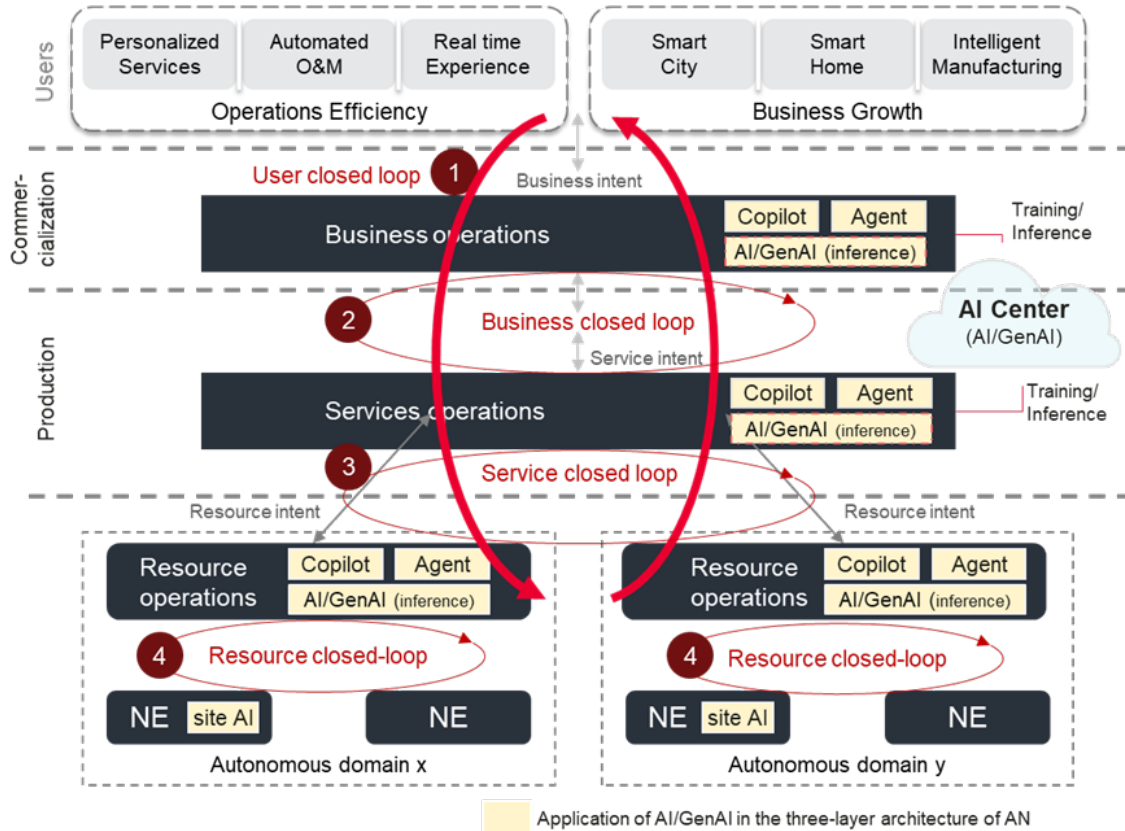


Figure 7. Full-stack AI/GenAI in the AN architecture

GenAI can be used at any operational layer in the AN reference architecture and can work in tandem with other types of AI to develop two primary kinds of agent applications: role oriented copilots and scenario-based agents capable of acting on behalf of users or systems. This integration significantly enhances autonomous capabilities across all operational layers (business, service, and resource operations). The AN Reference Architecture inherently distributes intelligence and knowledge across all three operational layers, with each autonomous domain containing intelligence specific to its functions. As AI continues to evolve, the reference architecture remains stable and future-proof, incorporating new AI capabilities without requiring structural changes.

2.4. AN Map / High-Value Scenarios

The shift towards network automation and intelligence is a complex transformation involving multiple organizations, diverse networks, and the cooperation of different operators' platforms, systems, and multi-vendor devices. Historically, network improvements often focused on isolated innovations rather than a holistic approach,

largely due to siloed departments and bottom-up methodologies. This fragmented approach hindered the streamlining of end-to-end network processes, ultimately slowing progress toward Autonomous Networks objectives.

To address these challenges, the "AN Map" was initially introduced as a tool to help Communication Service Providers define clear AN objectives, break them down into actionable steps, and prioritize capability development. The AN Map also provided guidance on planning and deployment scope while serving as a reference point for tracking AN evolution and successes. As the understanding of AN development has matured, so has the terminology. The AN Map has evolved into the High-Value Scenarios (HVS) Matrix, reflecting a more structured and scenario-driven approach. Rather than being just a static reference, this transformation represents an ongoing activity, namely, AN scenario mapping, which provides a way to dimension and organize high-value scenarios across different network lifecycle phases and service domains.

2.4.1. High-Value Scenarios and AN Scenario Mapping

To construct the HVS Matrix, we first define AN implementation scenarios, leveraging methodologies such as value stream analysis and frameworks like TOGAF [12]. This process involves analyzing 11 operational flows across services and networks, which are then integrated with service categories and network domains.

This structured approach forms the AN scenario framework, depicted in Figure 8, which helps define and prioritize AN high-value scenarios.

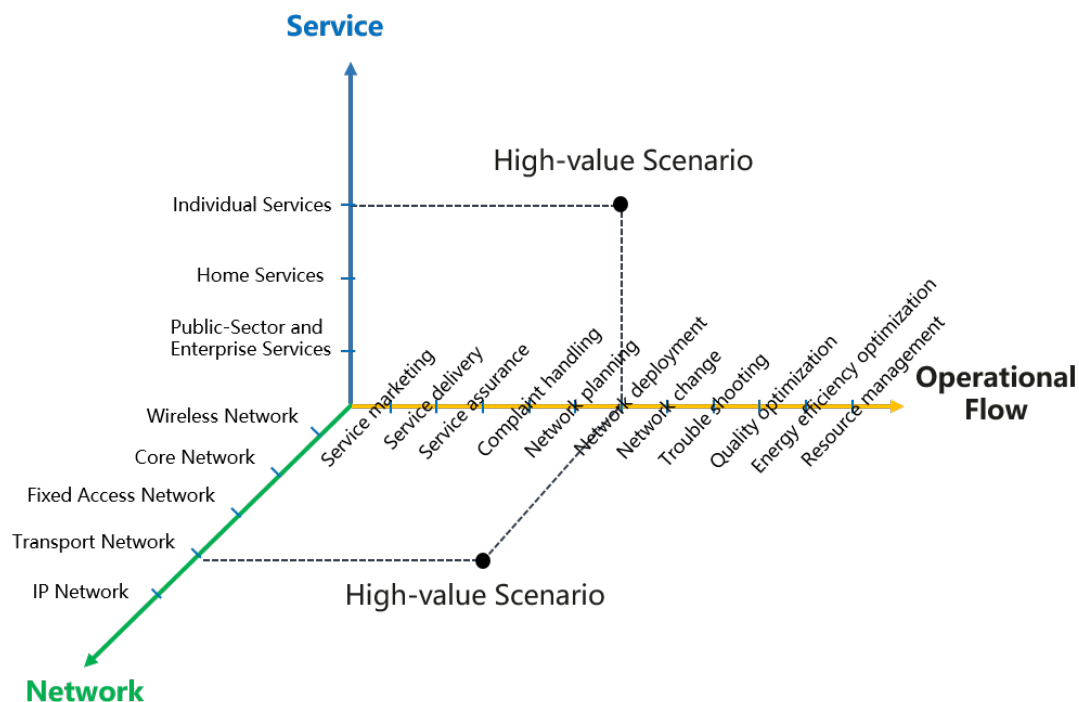


Figure 8. Mapping High-Value Scenarios Across Network, Service, and Operational Dimensions

2.4.2. Prioritize the Scenarios

The AN scenario framework, as discussed in the previous section, defines a comprehensive set of high-value implementation scenarios necessary for improving the AN level and ultimately attaining higher levels of autonomy from an end-to-end perspective. However, in practical application, multiple rounds of iteration and evolution are often necessary to make the most of limited resources and experience.

Hence, it is advisable to determine the priority of each AN high-value implementation scenario. This prioritization can be based on factors like the operator's service strategy, current autonomy status, and trends in service/network changes, as illustrated in Figure 9.

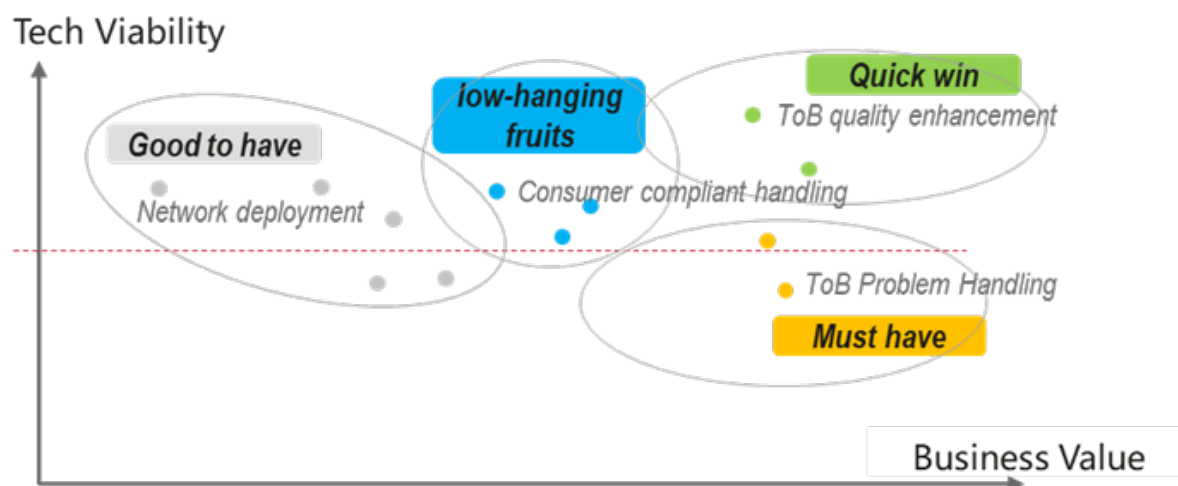


Figure 9. Four-quadrant analysis method for prioritize scenarios

The four-quadrant analysis method as a recommendation for prioritizing scenarios can be used to classify each high-level implementation scenario based on the assessment of business value and technology viability. Then the AN map is finally established, by giving priority to high-value implementation scenarios, practitioners can effectively concentrate limited resources to maximize the goal achievement within a single iteration.

Table 9 lists the AN L4 high-value scenarios in 2025-2027 described in L4 Blueprint [13].

AN L4 high-value scenarios in 2025-2027 (Reference)

Service-oriented		Individual Services	Home Services		Enterprise Services			
		4G/5G	Home Broadband	...	Private line	5G2B	IOT	...
Operati on	Service marketing	☐	☐	☐	☐	☐	☐	☐
	Service delivery	☐	☐	☐	☒	☒	☐	☐
	Service assurance	☒	☒	☐	☒	☒	☐	☐
	Complaint handling	☒	☒	☐	☐	☒	☐	☐
Network-oriented		Wireless	Core	Fixed access	Transport	IP	Others: e.g. Cloud	
Plan	Network planning	☐	☐	☐	☐	☐	☐	☐
	Network deployment	☐	☐	☐	☐	☐	☐	☐
Mainte nance	Fault management	☒	☒	☐	☒	☒	☐	☐
	Network change	☐	☒	☐	☒	☒	☐	☐
Optimi zation	Quality optimization	☒	☐	☐	☐	☒	☐	☐
	Energy efficiency optimization	☒	☐	☐	☒	☐	☐	☐
	Resource management	☐	☐	☐	☐	☐	☐	☐

☒ AN L4 high-value scenarios

Table 9. AN L4 High-Value Scenario mapping in 2025-2027 (Reference)

3. Autonomous Networks Practice

AN operational practices consist of two key activities: AN Strategic Planning (ANSP) and the AN Journey (refer to Figure 10).

- AN Strategic Planning provides a structured approach for CSPs (or subgroups within CSPs) to define and guide their AN evolution. It includes corporate-level AN strategies, executive commitments, and guidance on implementing the four foundational elements of the Autonomous Network Framework. Additionally,
- The AN Journey is an iterative process where operators implement strategies and continuously refine their approach to achieve their AN objectives. This distinction ensures that AN practices are understood as ongoing activities rather than static components.

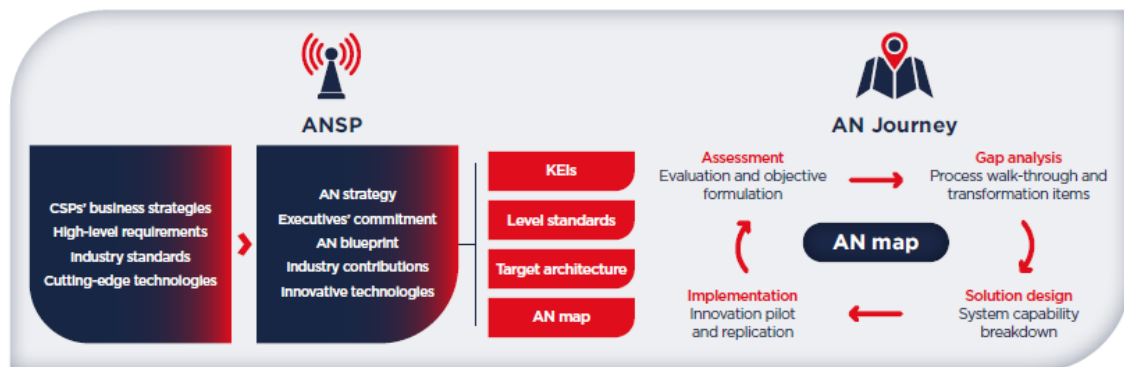


Figure 10. AN Operational Practices

To accelerate large-scale AN deployment, CSPs and their suppliers have devised a four-step continuous-improvement process based on high-value scenarios. By following this iterative process, operators can enhance their AN Level and effectiveness, and establish end-to-end service and business automation and autonomy.

1. **Assessment:** for each high-value scenario, CSPs evaluate the live network based on the AN Level and KEIs, establish baselines for domain-specific AN Level and KEIs, and set improvement goals based on service requirements and the investment plan.
2. **Gap analysis:** CSPs analyze the gaps between the baselines and goal, identify the root causes of breakpoints and weaknesses through process walk-through, break down the goal to achieve a higher AN Level and improve effectiveness, and formulate a key transformation item list.
3. **Solution design:** based on the target architecture, CSPs convert capability requirements into function requirements; design an application architecture; categorize system functions; output process transformation requirements and function requirements for OSSs, OMCs, and NEs; create system development lists; and pilot solutions.
4. **Implementation:** based on the system development list and pilot solutions, CSPs launch the solutions at opcos to facilitate the transition to production and replication among more opcos to improve their AN Level and KEIs and achieve the business goal.

4. Terms & Abbreviations Used within this Document

4.1. Terminology

Terminology used in this document is defined in IG1258 Autonomous Networks Glossary v1.1.0. [6]. Some additional terminology used in this guide is defined below.

Term	Definition
Autonomous Network Framework (ANF)	The TM Forum's Autonomous Network Framework is an implementation methodology designed to help CSPs plan and implement Autonomous Networks in an efficient and systematic way.
O&M	Operations and Maintenance
KCI	Key Capability Indicator
KPI	Key Performance Indicator
KEI	Key Effectiveness Indicator

5. References

- [1] [Autonomous Networks Business Requirements and Framework \(IG1218\) – TM Forum](#)
- [2] [Autonomous Networks – Reference Architecture \(IG1251\) – TM Forum](#)
- [3] [Autonomous Network Levels Evaluation Methodology \(IG1252\) – TM Forum](#)
- [4] [IG1256 Autonomous Network Effectiveness Indicators – TM Forum](#)
- [5] [IG1251A Autonomous Networks Reference Architecture Realizations – TM Forum](#)
- [6] [IG1258 Autonomous Networks Glossary – TM Forum](#)
- [7] [TS28.100 Management and orchestration; Levels of autonomous network, 3GPP](#)
- [8] [Autonomous Networks: Empowering digital transformation – from strategy to implementation \(IG1305\) – TM Forum](#)
- [9] [Autonomous Networks: Empowering digital transformation – evolving from Level 2/3 towards Level 4 \(IG1326\) – TM Forum](#)
- [10] [GB935 Metrics Framework R19.0.1](#)
- [11] [GB988 TM Forum Metrics Definitions v21.5 – TM Forum](#)
- [12] [The TOGAF Standard, 10th Edition, The Open Group](#)
- [13] [IG1401 TM Forum AN Journey Guide: Level 4 industry blueprint high-value scenarios](#)

6. Administrative Appendix

6.1. Document History

6.1.1. Version History

Version Number	Date Modified	Modified by:	Description of changes
0.9.0	01-Oct-2023	Kevin Xu, APAC Program Director, TM Forum Kevin McDonnell, Senior Director, Huawei	First Draft
1.0.0	17-Oct-2023	Alan Pope	Final edits prior to publication
1.1.0	04-Jul-2024	Alan Pope	Final edits prior to publication
1.2.0	10-Jan-2025	Alan Pope	Final edits prior to publication
2.0.0	14-Mar-2025	Yuan Yao, Wu Yunhe, Kevin McDonnell	Revisions to section 2.3

6.1.2. Release History

Release Status	Date Modified	Modified by:	Description of changes
Pre-production	17-Oct-2023	Alan Pope	Initial Release
Pre-production	20-Nov-2023	Adrienne Walcott	Updated to Member Evaluated status
Pre-production	04-Jul-2024	Alan Pope	Updated to v1.1.0
Production	30-Aug-2024	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	10-Jan-2025	Alan Pope	Updated to v1.2.0
Production	07-Mar-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status
Pre-production	14-Mar-2025	Alan Pope	Updated to v2.0.0
Production	09-May-2025	Adrienne Walcott	Updated to reflect TM Forum Approved status

6.2. Acknowledgments

This document was prepared by the members of the TM Forum Autonomous Networks team.

Team Member (@mention)	Company	Role*
Kevin Xu	TM Forum	Author
Kevin McDonnell	Huawei	Author, Editor, Project Chair
Hui Li	Huawei	Editor, Key Contributor
guangying zheng	Huawei	Editor, Key Contributor
Yuan Yao	China Mobile	Key Contributor
JUAN ANDRÉS SUELS CARDOZO	Telefónica	Key Contributor
Asser EL NAHAS	Orange	Key Contributor
Chao Wu	NTT	Key Contributor
Yunhe Wu	Huawei	Key Contributor
Xiao Hongmei (Malina)	Inspur	Reviewer
Dmytro Gassanov	TM Forum	Reviewer
Alan Pope	TM Forum	Reviewer, Editor

*Select from: Project Chair, Project Co-Chair, Author, Editor, Key Contributor, Additional Input, Reviewer